

State-Dependent Anonymity (Addendum): Calibration Vignettes and Time-to-Identification Conversions

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Abstract

This companion note collects short calibration vignettes for the State-Dependent Anonymity model (State 1). It is intentionally non-redundant: we do not restate the full game definitions or upper bounds. Instead we record two reusable translations: (i) converting a per-epoch witness gap into time-to-identification, and (ii) converting a per-lookup MaxL bound into a Boss-Fight horizon budget.

Scope. Calibration addendum for State 1.[1]

Units. Budgets are published in bits (use $\log_2$ and 2^ϵ); results stated in nats can be converted by dividing by $\ln 2$ (see Synthesis 8).[5]

Imports. We import the state-dependent anonymity viewpoint and witness taxonomy from State 1.[1] Endpoint sizing conventions and MaxL/PML vocabulary follow the endpoint bridge.[3] Receipt publication fields follow the receipt line-item schema.[4] Artifact availability follows the artifact policy.[6]

Exports. Simple conversions and small tables that are receipt-ready and help operators interpret “small” per-epoch gaps.

1 Gap-to-time conversion for repeated public witnesses

Definition 1 (Per-epoch witness gap). Let $W_t \in \{0, 1\}$ be a public witness bit observed once per epoch (reconfiguration, consensus, or lookup), with $\Pr[W_t = 1 \mid C = 1] = p_1$ and $\Pr[W_t = 1 \mid C = 0] = p_0$. Define the per-epoch gap $\Delta := |p_1 - p_0|$.

Theorem 1 (Rule-of-thumb identification horizon). *Fix target total error $\eta \in (0, 1)$. For i.i.d. Bernoulli witnesses with gap Δ , a simple likelihood-ratio test achieves error $\leq \eta$ once*

$$T \gtrsim \frac{2\log_2(1/\eta)}{\Delta^2}.$$

Conversely, any test requires $T = \Omega(\log_2(1/\eta)/\Delta^2)$ epochs when Δ is small.

Epoch-to-time. To translate epochs into wall-clock time, pick a witness cadence (per consensus period, per reconfiguration, or per lookup). Table 1 instantiates Theorem 1 at $\eta = 0.05$.

Gap Δ	Required epochs T	Time @ 5 min	Time @ 1 hour
0.01	6.0×10^4	≈ 208 days	≈ 6.8 years
0.02	1.5×10^4	≈ 52 days	≈ 1.7 years
0.05	2.4×10^3	≈ 8.3 days	≈ 100 days

Table 1: Epoch-to-time conversion for Theorem 1 at $\eta = 0.05$.

When epochs are “per lookup.” If the witness occurs once per lookup and a client issues r lookups/day, then T epochs corresponds to T/r days. This is one reason receipts often publish a horizon budget (queries per window) rather than a clock-time guarantee.

2 Per-lookup MaxL to Boss-Fight horizon

If a deployment declares a per-lookup MaxL bound of the form

$$\mathcal{L}_{\max}(C \rightarrow Y) \leq \varepsilon$$

for the declared projection Y , then under the conservative repeated-use accountant

$$\mathcal{L}_{\max}(C \rightarrow Y_{1:Q}) \leq Q \varepsilon.$$

This bound is tight in adversarial stop/abort settings unless additional independence or reset structure is declared.[2]

Receipt-ready line item. Publish (Q, ε) (or a tiered list (Q_i, ε_i)) using the receipt schema.[4]

References

- [1] Anonymous. State-Dependent Anonymity: Selection and State Evolution as an Attack Surface. AnonDHT State series Paper 1, 2026. (This archive.)
- [2] Anonymous. Boss Fight Budgets: Repeated-Use Identification Accounting for Anonymous DHT Lookups. Boss Fight series Paper A, 2026. (This archive.)
- [3] Anonymous. Endpoint Metrics Bridge: Odds Inflation, PML/MaxL, and Identification Sizing. Series synthesis note (Synthesis 5), 2026. (This archive.)
- [4] Anonymous. Receipt Line-Item Schema for Anonymous-DHT Privacy Claims. Series synthesis note (Synthesis 12), 2026. (This archive.)
- [5] Anonymous. Notation and Minimal Model for the One-True-Archive. Synthesis 8, The-One-True-Archive (2026).
- [6] Anonymous. Artifact policy and supporting materials for the anonymity/AnonDHT series. 2026.